

Future Interns · Cyber Security Track · Task 2

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1. EXECUTIVE SUMMARY

This Phishing Email Detection & Awareness Report documents the analysis of 10 email samples from public datasets (PhishTank, OpenPhish). Each email was examined for spoofed senders, deceptive URLs, social engineering, and malicious attachments. Results: **5 Phishing, 3 Suspicious, 2 Safe**.

Classification	Count	Action
PHISHING	5	Block sender, report to IT Security, do not click
SUSPICIOUS	3	Quarantine, verify via alternate channel
SAFE	2	No action required

2. ANALYSIS METHODOLOGY

Each email sample was analysed using a structured 5-step process:

- **Header Analysis:** Examined From, Reply-To, Return-Path for spoofing; verified DKIM/SPF/DMARC.
- **Sender Verification:** Compared display name with sending domain; checked domain age via WHOIS.
- **URL Analysis:** Hovered over all links; checked destinations against PhishTank and VirusTotal.
- **Content Analysis:** Identified urgency language, threats, grammar errors, brand impersonation.
- **Attachment Review:** Noted suspicious attachments (.exe, macro-enabled Office files, password ZIPs).

3. EMAIL SAMPLE ANALYSIS TABLE

#	Subject	Sender Domain	Verdict	Key Indicator
E-01	Your account has been suspended!	paypa1-secure.com	PHISHING	Typosquatted domain
E-02	Verify your identity – Netflix	netflix-billing.xyz	PHISHING	Fake domain + urgency
E-03	Invoice attached – Action needed	invoices-hr.ru	PHISHING	Malicious attachment
E-04	Update your password now	amazon-security.net	PHISHING	Credential harvesting
E-05	Your parcel could not be delivered	dhl-tracking.info	PHISHING	Fake tracking link
E-06	Unusual sign-in activity	accounts.google.com.sso.xyz	SUSPICIOUS	Subdomain spoofing
E-07	Meeting invitation – please confirm	calendar-noreply.net	SUSPICIOUS	Unverified sender
E-08	Your subscription renewal	noreply@company-billing.co	SUSPICIOUS	Unofficial domain
E-09	Monthly newsletter – June 2025	newsletter@trustedorg.com	SAFE	Verified sender, DKIM pass

#	Subject	Sender Domain	Verdict	Key Indicator
E-10	Project update from manager	manager@yourcompany.com	SAFE	Internal domain, SPF pass

4. DETAILED CASE STUDIES

E-01 – Your account has been suspended!

From	PayPal Security <service@paypa1-secure.com>	Verdict	PHISHING
Technique	Brand Impersonation + Typosquatting + Urgency		

Phishing Indicators:

- Domain 'paypa1-secure.com' uses number '1' to mimic letter 'l'
- Urgent threat: account deleted in 24 hours
- Link: http://paypa1-secure.com/verify (not paypal.com)
- Reply-To address differs from From address
- SPF record: FAIL

Business Impact: Credential theft leading to financial account takeover.

E-02 – Verify your identity – Netflix

From	Netflix Billing <billing@netflix-billing.xyz>	Verdict	PHISHING
Technique	Brand Impersonation + Payment Harvesting		

Phishing Indicators:

- 'netflix-billing.xyz' registered 3 days before email was sent
- Embedded button links to phishing kit on compromised server
- Email requests full credit card details including CVV
- No DKIM signature present
- HTML formatting inconsistent with Netflix brand

Business Impact: Financial data theft; credit card fraud.

E-03 – Invoice attached – Action needed

From	Accounts Payable <accounts@invoices-hr.ru>	Verdict	PHISHING
Technique	Malicious Attachment + Business Email Compromise		

Phishing Indicators:

- Attachment: Invoice_June2025.xlsm (macro-enabled Excel)
- Subject creates urgency around payment deadline
- Sender domain registered in Russia with no business connection

- VirusTotal: 14/68 engines flagged as Trojan.Macro
- Email body contains no company-specific details

Business Impact: Malware installation; ransomware deployment risk.

5. PHISHING INDICATOR REFERENCE

Category	Red Flag	Example
Sender	Domain typosquatting	paypa1.com vs paypal.com
Sender	Display name spoofing	'PayPal'
Sender	Free email for businesses	company_billing@gmail.com
URL	Domain mismatch	Shown: paypal.com → Actual: paypa1.xyz
URL	URL shortener used	bit.ly/3xyzABC – hides destination
URL	HTTP not HTTPS	http://login.mybank.com
Content	Urgency / threats	'Respond in 24 hrs or account deleted'
Content	Grammar/spelling errors	'Dear Valued Costumer'
Attachment	Macro-enabled Office file	.xlsm, .docm, .pptm
Headers	SPF/DKIM/DMARC fail	Email auth checks fail

6. USER AWARENESS GUIDELINES

Before You Click

- Hover over links before clicking to preview the actual URL.
- Verify sender email matches the company's official domain.
- Check for HTTPS and valid SSL on any login page.

Handling Suspicious Emails

- Do not open attachments from unknown or unexpected senders.
- Report suspicious emails to your IT/Security team immediately.
- Verify unusual payment/password requests via phone.

Technical Controls

- Enable Multi-Factor Authentication (MFA) on all accounts.
- Deploy DMARC, DKIM, and SPF on all company email domains.
- Implement email gateway filtering with real-time URL scanning.

7. CONCLUSION

Of 10 emails analysed, 50% were confirmed phishing and 30% suspicious. Technical controls and user education must work together to reduce risk. Report prepared as part of Future Interns Cyber Security Internship (Task 2) by Lokesh (GitHub: lokeshlls123456 | Repository: FUTURE_CS_02).